

Simulation-Based Appointment Scheduling for Urgent and Routine Healthcare Demand

MATH6185 | OPERATIONAL RESEARCH AND DATA SCIENCE

Case Study 1

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1. Introduction

Healthcare appointment systems must balance timely access for urgent patients against equitable access and efficient use of limited capacity. When demand is uncertain and short-run capacity is fixed, reserving appointments for urgent demand can shorten urgent waits but leave capacity unused and delay routine patients. Assigning every patient to the first available position avoids this loss of capacity but ignores clinical priority.

This report studies a simplified appointment-based healthcare operation with two patient classes. Urgent and routine patients arrive according to independent Poisson processes at rates 3.69 and 2.19 patients per working day. Each working day contains six equal appointment positions. Treatment times are deterministic and equal to one position, appointments are never missed or cancelled, and every patient receives an appointment immediately on arrival. Performance is measured by the time from arrival to the start of treatment, separately for urgent patients, routine patients and all patients.

Expected demand is 5.88 patients per day against a capacity of six, giving a nominal offered load of 0.98 and average spare capacity of only 0.12 positions per day. Small changes in protected capacity can therefore have a large effect on waiting times.

The report reviews mathematical approaches to multi-priority scheduling, establishes a First Available Slot (FAS) benchmark, and evaluates two alternatives: a workload-responsive rule that delays routine bookings, and an on-call routine (OCR) policy that releases protected near-term capacity to flexible routine patients. The policies are assessed through independent replications, warm-up checks and 95% confidence intervals, subject to an overall mean-wait constraint of five days.

2. Literature Review

2.1 Scope and decision hierarchy

The relevant operational problem is a stream of heterogeneous appointment requests competing for a resource whose ordinary capacity is fixed or expensive to expand. Each request belongs to a priority class, and the scheduler allocates current or future capacity while trading urgent access against routine waiting, idle time, late service and possibly overtime. This structure occurs in diagnostic imaging, primary and specialty care, elective services, public-sector appointments and repair systems.

Three features distinguish it from a conventional single-class appointment problem. First, a highly loaded resource has little slack, so protected capacity has a high opportunity cost. Second, classes have asymmetric delay consequences. Third, decisions are dynamic because the scheduler observes the current appointment book and backlog when each request arrives. Overtime changes the capacity

boundary, while cancellations, released protection or flexible customers create later opportunities to revise bookings.

Cayirli and Veral (2003) classify outpatient scheduling studies by environmental conditions and performance measures and emphasise the recurring trade-off between patient waiting, provider idle time and overtime. Gupta and Denton (2008) identify interacting decisions concerning capacity, appointment-book design, patient differentiation and real-time control. Ahmadi-Javid, Jalali and Klassen (2017) organise optimisation studies into strategic, tactical and operational levels. In the present problem, six daily positions are a fixed strategic input; the amount of capacity protected for urgent demand is a tactical choice; and the slot given or later offered to an individual patient is an operational decision.

2.2 Static allocation and mathematical programming

A common tactical formulation allocates a fixed number of positions to patient classes before demand is realised. If x_k is capacity allocated to class k , $W_k(x)$ is expected class waiting and w_k is a priority weight, a generic formulation is:

$$\begin{aligned} \min & \sum_k w_k E[W_k(x)] \\ \text{subject to} & \sum_k x_k \leq c, x_k \geq 0. \end{aligned}$$

The objective may instead minimise target violations, idle capacity or a weighted combination of waiting and overtime. Service-level constraints such as $P(W_k \leq d_k) \geq \beta_k$ can protect a class directly. If overtime o is available, it is usually penalised in the objective or restricted by $E[o] \leq o_{\max}$.

Creemers, Belien and Lambrecht (2012) combine a bulk-service queue with quadratic optimisation and heuristics to allocate server slots among patient classes. They find that utilisation generally must remain below 100% for waits to remain controlled and that spreading a class's positions across a cycle can reduce gaps between service opportunities. A contiguous protected block may shorten urgent access but create long gaps for routine patients.

Dobson, Hasija and Pinker (2011) study reserved urgent capacity in primary care. Their stochastic formulation balances urgent demand that cannot be handled during normal hours against the non-urgent queue and the cost of unused protection. Complete separation is inefficient under variable demand when unused urgent positions cannot be filled by routine demand. The value of reservation depends on demand variability, delay costs and whether lower-priority customers can use capacity that would otherwise expire.

Static allocation models are transparent and often easy to implement, but they reduce the system to forecasts or a small scenario set. They cannot fully exploit the exact pattern of free positions, patient ages and information revealed during the booking process.

2.3 Queueing and Markov models

Queueing formulations represent class demand as stochastic arrivals and service as discrete or continuous capacity. A discrete-time bulk-service representation is:

$$Q_{(k,t+1)} = \max(0, Q_{(k,t)} - S_{(k,t)}) + A_{(k,t)},$$

with $\sum_k S_{(k,t)} \leq c_t + o_t$. Here $Q_{(k,t)}$ is the class backlog, $A_{(k,t)}$ is new demand, $S_{(k,t)}$ is allocated service and o_t is overtime. A long-run or discounted objective may penalize $\sum_k h_k Q_{(k,t)} + p o_t$, with larger holding cost h_k for more urgent classes.

Izady (2015) uses discrete-time queues for appointment-capacity planning in specialty clinics, distinguishing rapid-access urgent demand from non-urgent patients booked into future sessions. The model links capacity to access targets while representing backlogs and operational features such as cancellations, no-shows and rescheduling. Average capacity can be sufficient while variability in demand and service availability still causes access-target failures.

Queueing models make the stability condition explicit: long-run effective demand must be below effective capacity. Here the offered load is $5.88/6 = 0.98$ before any inefficient protection. The small slack explains why a policy that permanently withholds too many routine positions can create severe delay even though total mean demand is below capacity.

When each request is scheduled after observing the booking profile, a Markov decision process (MDP) is natural. Its state may contain class-specific bookings on future days, waiting-list ages and available ordinary or overtime capacity. The action assigns an appointment, defers or rejects demand, or activates overtime. The Bellman equation balances immediate priority-weighted delay and capacity cost against the expected future value of flexibility.

Patrick, Puterman and Queyranne (2008) formulate dynamic multipriority diagnostic scheduling as an MDP with several wait-time targets and variable capacity. Because the exact state space is too large, they use approximate dynamic programming and linear-programming approximations. Their results show that differentiated targets can be achieved without fully dedicated resources and that the future booking profile matters, not only the next free date. The principal disadvantage is dimensionality: every added future period, class and capacity state enlarges the model, so aggregation and simulation are required in practice.

2.4 Online scheduling and overtime

Online models require fewer assumptions about the arrival sequence. Keyvanshokoo, Shi and Van Oyen (2021) consider heterogeneous requests that must be assigned to finite provider capacity, with overtime available to expand capacity. Their primal-dual algorithms use capacity opportunity costs

and provide worst-case performance guarantees. The approach performs well against first-come, first-served and threshold benchmarks on operational data.

This approach values a position according to its scarcity and the future demand that may need it. Higher-priority demand can face a lower effective booking cost, while overtime is used only when its benefit exceeds its penalty. Online methods are useful when a precise stationary arrival model is doubtful, although they do not directly provide the stochastic confidence intervals required here.

2.5 Simulation optimisation and practical policy structures

Discrete-event simulation can retain the exact appointment book, sequential arrivals, class rules, rebooking and stopping condition. Simulation optimisation evaluates a parameterised family of implementable policies. The Q3 problem has the form:

$$\begin{aligned} \min_{\theta} E[W_U(\theta)] \\ \text{subject to } E[W(\theta)] \leq 5. \end{aligned}$$

Simulation is flexible, but it does not identify an optimum by itself. It requires a defensible parameter search, independent replications, a warm-up rule and confidence intervals. Replication means, rather than individual patient observations, should be used for confidence intervals because observations within one run are serially dependent.

Five policy structures recur in the literature:

1. Dedicated or reserved capacity protects urgent demand but may waste positions when urgent demand is low.
2. Nested booking limits allow high-priority classes to access more capacity than low-priority classes without fully separating them.
3. Priority-weighted workload penalties postpone lower-priority bookings according to the observed high-priority backlog.
4. Release, call-in and rebooking rules offer protected positions to flexible lower-priority patients when the urgent value of the capacity has decayed.
5. Overtime thresholds activate surge capacity when backlog, lateness or marginal delay cost exceeds a limit.

Patrick and Puterman (2007) provide the closest precedent for Q4. Their diagnostic service has high-priority inpatients and lower-priority outpatients. They preserve capacity for urgent demand but call suitable outpatients into positions that would otherwise remain unused. Their optimisation minimises unused capacity subject to an overtime constraint, and simulation shows that flexible use can control outpatient delay without abandoning priority. The mechanism is not simply to give routine patients

more capacity; it is to preserve a short urgent-protection window and recycle capacity whose urgent value has diminished.

These studies support the three policies evaluated below. FAS provides a pooled benchmark. Q3 uses a convex urgent-workload penalty as a state-dependent approximation to an MDP value function, while Q4 combines initial reservation with next-day release to OCR patients. With a load of 0.98, protection must be limited and unused appointments released promptly.

3. Methods

3.1 System and common assumptions

Urgent and routine arrivals are independent Poisson processes with daily rates 3.69 and 2.19. Time is measured in working days. Each day has six equally spaced positions, and treatment takes exactly one position. There are no no-shows, cancellations, patient preferences, class changes or non-working periods. An initial appointment is made immediately at arrival. Rebooking is allowed only for OCR patients in Q4.

The earliest possible appointment for an arrival is the first position that begins after the arrival time. Each policy then searches the future appointment book according to its rule. Waiting time is appointment start time minus arrival time. The simulation stops immediately after the 50,000th arrival and does not process the remaining appointments.

The benchmark warm-up condition is 1,000 patients served. Because the appointment time and therefore the wait are known when a booking is made, a new observation is admitted once 1,000 treatments have begun. This avoids censoring patients who have an appointment but have not started treatment when the 50,000th arrival occurs. In Q4, a rebooking reduces the recorded total by the precise amount of time saved.

3.2 Policies

First Available Slot

FAS assigns every arrival to the earliest unbooked position after arrival. It does not use patient class. Consequently, its urgent and routine long-run means should be equal; separate class statistics provide a validation check.

Workload-responsive priority policy

Urgent patients continue to receive FAS. For a routine arrival, the scheduler first identifies the position that FAS would provide. It then postpones the search by

$$p = \text{round}(\alpha_1 N_U + \alpha_2 N_U^2)$$

positions, where N_U is the number of urgent patients currently waiting for treatment. Starting from the postponed position, the scheduler scans forward to the first vacancy. N_U increases when an urgent patient arrives and decreases when that patient starts treatment.

The rule protects more future capacity when urgent workload is high. The quadratic term makes protection grow rapidly in congested states while leaving routine patients largely unaffected when few urgent patients are waiting. The coefficients were searched by pilot runs and then evaluated through formal replications.

On-call routine policy

The OCR policy is adapted from Patrick and Puterman (2007) and uses integers $0 \leq L < H < 6$. Urgent arrivals receive FAS. Routine arrivals use a modified FAS rule under which the first H positions of each day are unavailable during the initial booking.

At the end of each day, the next day's schedule is reviewed. The first L positions remain protected for urgent demand. Each other open next-day position is offered to the longest-waiting eligible OCR patient whose current appointment is later. A successful move releases the old appointment, books the earlier position and shortens that patient's recorded wait. OCR status is independently assigned to a fraction η of routine arrivals. The required η values 0.1 and 0.2 were evaluated.

3.3 Experimental design and estimation

Each formal policy estimate uses 250 independent replications and a distinct random seed per replication. If $W_{bar}ar_r$ is the post-warm-up mean from replication r , the estimator is the average of the replication means. A 95% confidence interval is:

$$\hat{\mu} \pm t_{(0.975, R-1)} \frac{s(\bar{W})}{\sqrt{R}}.$$

The reported relative precision is the confidence-interval half-width divided by the point estimate. The target is no more than 5%. Statistical feasibility for the overall five-day constraint is judged primarily from the point estimate, with the confidence-interval upper limit also reported as a conservative check.

All three selected models were run directly in AnyLogic Personal Learning Edition for 250 independent replications. Q3 also evaluated α_2 values 0.38 and 0.45 with α_1 fixed at zero. For Q4, integer threshold pairs for $\eta = 0.1$ and 0.2 were first screened with a reproducible event-equivalent runner. The selected policy was then validated directly in the submitted AnyLogic model using the same warm-up, rebooking and arrival-based stopping rules.

4. Results

4.1 Q2: First Available Slot benchmark

With the benchmark 1,000-served warm-up, the overall mean is 4.148 days and the 95% confidence interval is [3.978, 4.318]. Its relative half-width is 4.10%, meeting the precision target, and the assignment benchmark of approximately four days lies inside the interval. The urgent and routine means are 4.148 days in both cases, as expected from a class-blind policy.

Table 1. FAS warm-up assessment, 250 independent replications.

Treatments started before collection	Mean wait (days)	95% confidence interval (days)	Relative half-width
500	4.133	[3.965, 4.302]	4.08%
1,000	4.148	[3.978, 4.318]	4.10%
2,000	4.172	[3.999, 4.345]	4.15%

Across the tested warm-ups, means differed by at most 0.038 days, versus confidence-interval half-widths of 0.169-0.173 days. The 1,000-patient benchmark was retained.

4.2 Q3: Workload-responsive priority

With α_1 fixed at zero, pilot runs identified three α_2 values around the five-day feasibility boundary.

Table 2. Workload-responsive policy results, 250 independent replications per candidate.

α_1	α_2	Urgent mean, days (95% CI)	Routine mean, days (95% CI)	Overall mean, days (95% CI)	Feasible?
0	0.35	2.123 [2.075, 2.172]	9.053 [8.621, 9.484]	4.703 [4.513, 4.893]	Yes
0	0.38	2.173 [2.117, 2.229]	10.019 [9.500, 10.538]	5.096 [4.868, 5.324]	No
0	0.45	2.118 [2.067, 2.169]	10.926 [10.385, 11.468]	5.397 [5.164, 5.630]	No

The selected 0.35 row is the direct AnyLogic PLE result. The other rows retain the estimates from the earlier formal candidate experiments.

The selected coefficients are $(\alpha_1, \alpha_2) = (0, 0.35)$. Relative to FAS, urgent mean waiting falls from 4.148 to 2.123 days, a reduction of 2.024 days or 48.8%. Routine mean waiting rises to 9.053 days.

The overall mean is 4.703 days, its confidence-interval upper limit is 4.893 days, and its relative half-width is 4.04%.

The urgent point estimates are not monotone across the three candidate experiments, and their confidence intervals overlap. Routine and overall waiting nevertheless increase once α_2 moves beyond the feasible boundary. With little spare capacity, postponing routine bookings creates vacancies for future urgent arrivals at the cost of a long routine tail.

4.3 Q4: On-call routine patients

Threshold screening showed that $H = 4$ was too aggressive. Even with $L = 0$, pilot overall means were about 20 days for $\eta = 0.1$ and 6.7 days for $\eta = 0.2$. Reserving four of six positions initially leaves too little effective routine capacity at a 0.98 load. The final comparison therefore focused on $H = 2$ and $H = 3$ with $L = 0$.

Table 3. OCR screening estimates and direct validation of the selected policy, 250 independent replications per policy.

η	L	H	Urgent mean, days (95% CI)	Routine mean, days (95% CI)	Overall mean, days (95% CI)	Slot utilisation
0.1	0	3	4.107 [3.967, 4.248]	4.691 [4.549, 4.832]	4.325 [4.183, 4.466]	97.936%
0.1	0	2	4.109 [3.968, 4.251]	4.323 [4.181, 4.464]	4.189 [4.047, 4.330]	97.939%
0.2	0	3	4.009 [3.862, 4.156]	4.574 [4.426, 4.722]	4.219 [4.072, 4.367]	97.911%
0.2	0	2	4.093 [3.952, 4.234]	4.304 [4.162, 4.446]	4.172 [4.030, 4.313]	97.939%

The third row is the direct AnyLogic PLE validation. The remaining rows are screening estimates from the event-equivalent runner.

All four estimates satisfy the five-day constraint, including when feasibility is assessed by the upper confidence limit. The directly validated selected policy has an overall relative half-width of 3.49%.

The selected OCR policy is $(\eta, L, H) = (0.2, 0, 3)$. Its direct AnyLogic estimate gives an urgent mean of 4.009 days with 95% confidence interval [3.862, 4.156], a routine mean of 4.574 days [4.426, 4.722], and an overall mean of 4.219 days [4.072, 4.367]. Appointment-slot utilisation is 97.911% [97.858%, 97.965%]. A rebooked OCR patient saves 0.515 days on average.

The event-equivalent screening estimate for $H = 2$ has a similar urgent mean and lower routine waiting. Because this comparison mixes screening-runner and direct AnyLogic estimates, it is not used as a formal paired comparison. $H = 3$ remains the submitted choice under the stated urgent-wait objective.

4.4 Cross-policy comparison

Table 4. Performance of the selected policies.

Policy	Parameters	Urgent mean (days)	Routine mean (days)	Overall mean (days)	Main interpretation
FAS	none	4.148	4.148	4.148	Pooled benchmark; no priority differentiation
Workload-responsive	$\alpha_1 = 0, \alpha_2 = 0.35$	2.123	9.053	4.703	Large urgent improvement, paid for by routine delay
OCR	$\eta = 0.2, L = 0, H = 3$	4.009	4.574	4.219	Feasible release policy; modest urgent improvement

Q3 gives the largest reduction in urgent waiting: 48.8% relative to FAS, while its overall confidence-interval upper limit remains below five days. Q4 reduces urgent waiting by 0.139 days, or 3.3%. Its release rule creates short-notice opportunities for OCR patients, but FAS already uses nearly all available capacity.

5. Conclusions

The workload-responsive policy, $(\alpha_1, \alpha_2) = (0, 0.35)$, performs best against the stated objective. It reduces urgent mean waiting from 4.148 to 2.123 days while keeping the overall mean at 4.703 days. Routine mean waiting rises to 9.053 days, so feasibility on the aggregate measure comes with a marked distributional cost.

The selected OCR policy, $(\eta, L, H) = (0.2, 0, 3)$, is also feasible. It recovers near-term positions for flexible routine patients, but its urgent mean of 4.009 days is only 3.3% below FAS. In this model, the release mechanism preserves capacity use but does not match the urgent-wait improvement achieved by Q3.

The findings depend on stationary Poisson demand, deterministic service, identical working days and the absence of no-shows, cancellations, preferences, clinical deterioration and overtime. The finite parameter searches do not prove global optimality. More importantly, the five-day aggregate constraint permits a long routine wait. An operational specification should therefore include a routine service-level constraint or an explicit weighting of class delays. Common random numbers would improve close policy comparisons; further work could combine the Q3 workload signal with Q4 release or adapt the protection thresholds to the observed backlog.

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